Lenses and image formation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use ray directions to explain converging and diverging lens behavior.

SUCCESS CRITERIA - I can...

1. Distinguish converging and diverging lenses.
2. Trace principal rays with a clear optical axis.
3. State whether an image is real or virtual in the given case.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A source approaching a stationary listener at **15 m/s** emits **500 Hz**.
1. Use v_{sound}=**340** and $f_{obs}=f_s\ v/(v-v_s)$.
2. **Find observed frequency**.

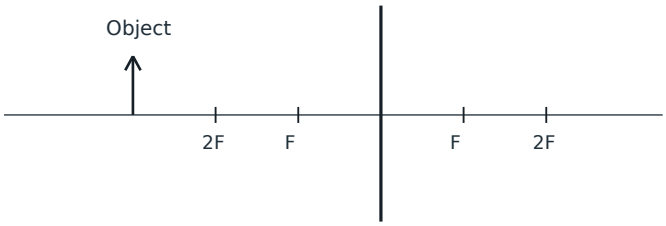
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\mathbf{1}/f = \mathbf{1}/d_o + \mathbf{1}/d_i$.
Thin lens: f is focal length; o = object, i = image. Use the stated sign convention and a ray diagram.

Worked reasoning: A converging lens can form a real image of an object beyond its focal point. A ray parallel to the axis passes through the far focus; a central ray is nearly straight when the lens is treated as very thin.

Your turn: Draw these two rays and **mark their intersection** for an object beyond **2** f.



Converging thin lens; add principal rays.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Complete ray sketches for parallel rays through a converging lens and a diverging lens. **Label the optical axis 0**.

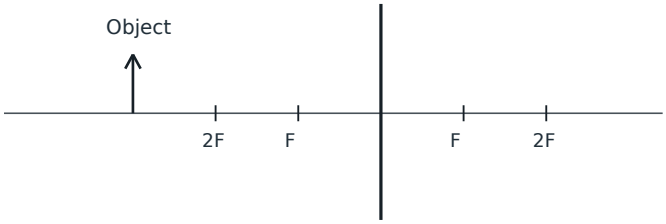


Left: converging lens. Right: diverging lens. Horizontal lines mark the optical axes.

Marzano check - circle: ① ② ③

04 Apply the physics

A converging lens has focal length **10 cm**. A small lamp is **30 cm** from it.
1. **Label the diagram**, find image distance using $\mathbf{1}/f = \mathbf{1}/d_o + \mathbf{1}/d_i$, and **calculate m** $= -d_i/d_o$.
2. **Check the sign** and size with two principal rays.



Converging thin lens; add principal rays.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A magnifying glass produces an upright virtual image when object is inside f. **Explain why** “converging lens always makes a real image” is wrong.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wave snapshot has crests at **0.2**, **0.6**, **1 m**. **Find wavelength**. Its period is **0.06 s**; **find frequency**. **Label a wave sketch** to **support the calculation or comparison**.

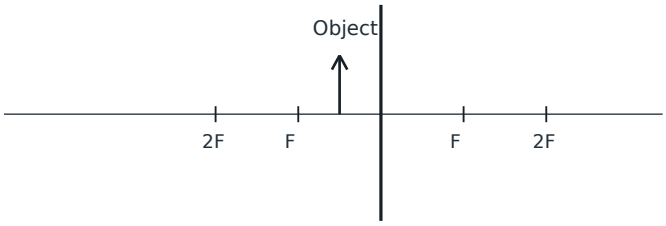


Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

For a converging lens f=**10 cm** and d_o=**5 cm**, find d_i and m. **State image kind**.

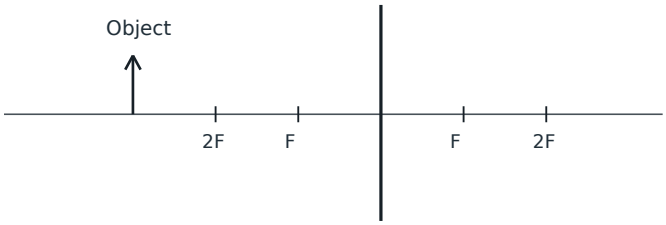


Converging thin lens; add principal rays.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A diverging lens has f = **-10 cm** and an object **30 cm** in front.
1. Use **1**/f = **1**/d_o + **1**/d_i and m = -d_i/d_o.
2. **Find image distance** and magnification, then **check them** with a ray sketch.



Diverging thin lens; add principal rays.

Marzano check - circle: 1 2 3